

作业纸

课程名称: 复变

1120212601

班级:

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1. 证: $f(z) = \begin{cases} \frac{\operatorname{Im}(z) \cdot \operatorname{Re}(z)}{|z|^2} & z \neq 0 \\ 1 & z = 0 \end{cases}$ $z = x + yi$

且 $\frac{\operatorname{Im}(z) \cdot \operatorname{Re}(z)}{|z|^2} = \frac{xy}{x^2 + y^2} = \frac{xy}{x^2 + y^2}$

令 $y = kx$ 的方向趋向

$\therefore \lim_{z \rightarrow 0} \frac{\operatorname{Im}(z) \cdot \operatorname{Re}(z)}{|z|^2} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{x^2 + y^2} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{kx^2}{(1+k^2)x^2} = \frac{k}{1+k^2} \neq 1$

其取值与 k 的取值有关.

$\therefore f(z) = \begin{cases} \frac{\operatorname{Im}(z) \operatorname{Re}(z)}{|z|^2} & z \neq 0 \\ 1 & z = 0 \end{cases}$ 在原点处不连续.

2. 证: $w = \frac{1}{z}$ 且令 $w = u + vi$

$\therefore z = \frac{1}{w} = \frac{1}{u+vi} = \frac{u-vi}{u^2+v^2} = x + yi$

$\therefore x = \frac{u}{u^2+v^2} \quad y = \frac{-v}{u^2+v^2}$

$\therefore$ 曲线 $x=2 \Rightarrow \frac{u}{u^2+v^2} = 2$

$\therefore 2u^2 + 2v^2 - u = 0$

$\therefore (u - \frac{1}{4})^2 + v^2 = \frac{1}{16} \Rightarrow |w - \frac{1}{4}| = \frac{1}{4}$



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$$3. \text{证: } f(z) = \frac{x}{x^2+y^2} - \frac{yi}{x^2+y^2}.$$

$$\therefore u(x,y) = \frac{x}{x^2+y^2}$$

$$v(x,y) = \frac{-y}{x^2+y^2}.$$

$$\therefore \frac{\partial u}{\partial x} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$\frac{\partial v}{\partial x} = \frac{2xy}{(x^2+y^2)^2}$$

$$\frac{\partial u}{\partial y} = \frac{-2xy}{(x^2+y^2)^2}$$

$$\frac{\partial v}{\partial y} = \frac{y^2-x^2}{(x^2+y^2)^2}.$$

$\therefore$ 由 C-R 方程得:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{且} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \quad \text{且} \quad x^2+y^2 \neq 0.$$

$\therefore f(z)$ 在除原点外的全平面上可导.

$$f'(z) = \frac{y^2-x^2}{(x^2+y^2)^2} + i \frac{2xy}{(x^2+y^2)^2} = -\frac{1}{z^2}$$

$\therefore f(z)$ 在除原点外的全平面上处处解析.



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4. 证: (i) 无计算 $\int_{|z|=1} \frac{e^z}{z} dz$

$$\because e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

$$\text{则 } \frac{1}{z} e^z = \sum_{n=0}^{\infty} \frac{z^{n-1}}{n!}$$

$$\therefore \text{令 } n-1 = -1 \Rightarrow n=0$$

$$\text{则 } C_{-1} = \frac{1}{0!} = 1$$

$$\therefore \int_{|z|=1} \frac{e^z}{z} dz = 2\pi i C_{-1} = 2\pi i$$

(ii) $\int_{|z|=1} \frac{e^z}{z} dz = 2\pi i$ 且 $z = e^{i\theta} \quad dz = ie^{i\theta} d\theta$

$$\therefore \int_0^{2\pi} \frac{e^{e^{i\theta}}}{e^{i\theta}} i e^{i\theta} d\theta = i \int_0^{2\pi} e^{e^{i\theta}} d\theta = 2\pi i \quad (\text{由 (i) 得})$$

$$\therefore i \int_0^{2\pi} e^{\cos\theta + i\sin\theta} d\theta = i \int_0^{2\pi} e^{\cos\theta} [\cos(\sin\theta) + i\sin(\sin\theta)] d\theta$$

$$\therefore i \int_0^{2\pi} e^{\cos\theta} [\cos(\sin\theta) + i\sin(\sin\theta)] d\theta = 2\pi i$$

$$\therefore \int_0^{2\pi} e^{\cos\theta} [\cos(\sin\theta) + i\sin(\sin\theta)] d\theta = 2\pi$$

由于积分是实部

$$\therefore \int_0^{2\pi} e^{\cos\theta} \cos(\sin\theta) d\theta = 2\pi$$

$$\therefore \int_0^{\pi} e^{\cos\theta} \cos(\sin\theta) d\theta = \frac{1}{2} \int_0^{2\pi} e^{\cos\theta} \cos(\sin\theta) d\theta = \pi$$

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5. 设 $2u(x,y) - v(x,y) = \frac{2x+y}{x^2+y^2}$

$$\therefore 2 \frac{\partial u}{\partial x} - \frac{\partial v}{\partial x} = \frac{2y^2 - 2x^2 - 2xy}{(x^2+y^2)^2} \quad (1)$$

$$2 \frac{\partial u}{\partial y} - \frac{\partial v}{\partial y} = \frac{x^2 - y^2 - 4xy}{(x^2+y^2)^2} \quad (2)$$

$$\text{且} \because \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\therefore 2 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = \frac{2y^2 - 2x^2 - 2xy}{(x^2+y^2)^2} \quad (3)$$

$$-\frac{\partial u}{\partial x} + 2 \frac{\partial u}{\partial y} = \frac{x^2 - y^2 - 4xy}{(x^2+y^2)^2} \quad (4)$$

$$\Rightarrow \begin{cases} \frac{\partial u}{\partial x} = \frac{y^2 - x^2}{(x^2+y^2)^2} \\ \frac{\partial u}{\partial y} = \frac{-2xy}{(x^2+y^2)^2} \end{cases}$$

$\therefore$ (i) 求 $u(x,y)$.

$$\int \frac{y^2 - x^2}{(x^2+y^2)^2} dx = \frac{x}{(x^2+y^2)} + C_1 y + C$$

$$\Rightarrow u(x,y) = \frac{x}{x^2+y^2} + C_1$$

$$\int \frac{-2xy}{(x^2+y^2)^2} dy = \frac{x}{(x^2+y^2)} + C_1 x + C$$

(ii) 求 $v(x,y)$ 且 $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

$$\therefore \int \frac{y^2 - x^2}{(x^2+y^2)^2} dy = \frac{-y}{x^2+y^2} + C_1 x + C$$

$$\int \frac{2xy}{(x^2+y^2)^2} dx = \frac{-y}{x^2+y^2} + C_2 y + C$$

$$\Rightarrow v(x,y) = \frac{-y}{x^2+y^2} + C_2$$

且已知 $2u(x,y) - v(x,y) = \frac{2x+y}{x^2+y^2} \Rightarrow 2C_1 - C_2 = 0 \Rightarrow C_2 = 2C_1$

$$\therefore f(z) = \frac{x}{x^2+y^2} + C_1 + i \left(\frac{-y}{x^2+y^2} + 2C_1 \right)$$

$$f(z) = \frac{1}{z} + C_1 + 2iC_1 = \frac{1}{z} + C_1 (1+2i)$$

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$$b. \text{ 证: } I = \oint_{|z|=1} \frac{|dz|}{|2z-1|^2} \quad \because |dz| = \frac{dz}{iz}$$

$$\therefore I = \oint_{|z|=1} \frac{1}{(2z-1)(2\bar{z}-1)} \frac{dz}{iz}$$

$$\therefore I = \oint_{|z|=1} \frac{z}{(2z-1)(2-\bar{z})} \frac{dz}{iz}$$

$$\therefore I = \oint_{|z|=1} \frac{\frac{1}{z}}{2(z-\frac{1}{2})(2-\bar{z})} dz \quad \text{其中 } z = \frac{1}{2} \text{ 为一极点}$$

$$\therefore \text{Res}(f(z), \frac{1}{2}) = \lim_{z \rightarrow \frac{1}{2}} \frac{\frac{1}{z}}{2(2-\bar{z})} = \frac{1}{3i}$$

$$\therefore I = 2\pi i \text{Res}(f(z), \frac{1}{2}) = \frac{2\pi}{3}$$

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$$7. \text{解: } f(z) = \frac{z}{(z+1)(z+2)} = \frac{A}{z+1} + \frac{B}{z+2} \Rightarrow \begin{cases} A+B=1 \\ 2A+B=0 \end{cases} \Rightarrow \begin{cases} A=-1 \\ B=2 \end{cases}$$

$$\therefore f(z) = \frac{2}{z+2} + \frac{(-1)}{z+1}$$

$$(i) \frac{2}{z+2} = \frac{2}{4+z-2} = \frac{\frac{1}{2}}{1+\frac{(z-2)}{4}} = \sum_{n=0}^{\infty} \frac{(-1)^n (z-2)^n}{2^{2n+1}} \quad R=4$$

$$(ii) \frac{-1}{z+1} = \frac{-1}{3+z-2} = \frac{-\frac{1}{3}}{1+\frac{(z-2)}{3}} = \sum_{n=0}^{\infty} \left(-\frac{1}{3}\right) \frac{(-1)^n (z-2)^n}{3^n} \quad R=3$$

$$\frac{-1}{z+1} = \sum_{n=0}^{\infty} \frac{(-1)^{n+1} (z-2)^n}{3^{n+1}}$$

$$\therefore f(z) = \sum_{n=0}^{\infty} (-1)^n (z-2)^n \left[\frac{1}{2^{2n+1}} - \frac{1}{3^{n+1}} \right]$$

其中 $R=3$ ($f(z)$ 的收敛半径).

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$$8. \text{ 证: } \sin^2 \frac{1}{z+1} = \frac{1}{2} - \frac{1}{2} \cos \frac{2}{z+1}.$$

当 $0 < |z+1| < \infty$ 时.

$$f(z) = (z+1)^{10} \left(\frac{1}{2} - \cos \frac{2}{z+1} \right) = \frac{1}{2} (z+1)^{10} - \frac{1}{2} (z+1)^{10} \cos \left(\frac{2}{z+1} \right).$$

$$\text{且 } \cos \left(\frac{2}{z+1} \right) = \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n} (z+1)^{-2n}}{(2n)!}.$$

$$\therefore f(z) = \frac{(z+1)^{10}}{2} \sum_{n=0}^{\infty} \frac{(-1)^n 4^n (z+1)^{-2n}}{(2n)!} + \frac{1}{2} (z+1)^{10}.$$

$$f(z) = \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n 4^n (z+1)^{-2n+10}}{(2n)!} + \frac{1}{2} (z+1)^{10}.$$

$\therefore -2n+10 = -1 \quad 2n = 11 \Rightarrow$ 不存在 $(z+1)^{-1}$ 项

$\therefore$ 此时沿正向圆周 $z=b$ 的积分 $= 0$.

(也可以说 $\frac{1}{2} - \frac{1}{2} \cos \frac{2}{z+1}$ 为偶函数, 不存在负一次幂).

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9. 解: $\int_{|z|=\frac{5}{2}} \frac{1}{(z-3)^2(z^5-1)} dz.$

解: 在 $|z|=\frac{5}{2}$ 内的极点有 $z = e^{i(\frac{2k\pi}{5})}$

在 $|z|=\frac{5}{2}$ 外的极点有 $z=3$ (2级极点) 与 $z=\infty$. 两个

$$\sum \text{Res} + \text{Res}(f(z), 3) + \text{Res}(f(z), \infty) = 0.$$

由 $\sum \text{Res} + \text{Res}(f(z), 3) = \text{Res}(\frac{1}{z^5} f(z), 0)$

(i) $\therefore \text{Res}(f(z), 3) = \frac{1}{1} \left(\frac{1}{z^5-1} \right)' \Big|_{z=3} = \frac{-405}{58564}$

(ii). $\int_{|z|=\frac{5}{2}} \frac{1}{z^5} \frac{1}{(z-3)^2(z^5-1)} dz = \int_{|z|=\frac{5}{2}} \frac{z^5}{(1-3z)^2(1-z^5)} dz$

$\therefore \text{Res}(\frac{1}{z^5} f(z), 0) = 0$

$\therefore \sum \text{Res} = 0 - \text{Res}(f(z), 3) = \frac{405}{58564} \times 2\pi i = \frac{405\pi i}{29282}$

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$$10 \int_0^{+\infty} \frac{4\cos x}{(x^2+1)(x^2+9)} dx.$$

解: 由使用留数计算定积分:

$$\int_{-\infty}^{+\infty} \frac{4\cos x}{(x^2+1)(x^2+9)} dx = \operatorname{Re} \left[\oint_C \frac{4e^{iz}}{(z^2+1)(z^2+9)} dz \right]$$

$$= \oint_C \frac{4e^{iz}}{(z+i)(z-i)(z+3i)(z-3i)} dz.$$

且 $z=i$ 与 $z=3i$ 在上半平面的极点.

$$\therefore \operatorname{Res}(f(z), i) = \frac{4e^{-1}}{(2i)(4i)(-2i)} = \frac{e^{-1}}{4i}$$

$$\operatorname{Res}(f(z), 3i) = \frac{4e^{-3}}{(4i)(2i)(6i)} = \frac{e^{-3}}{-12i}$$

$$\therefore \oint_C \frac{4e^{iz}}{(z+i)(z-i)(z+3i)(z-3i)} dz = \left(\frac{e^{-1}}{4i} - \frac{e^{-3}}{12i} \right) 2\pi i$$

$$= \frac{2\pi}{4e} - \frac{2\pi}{12e^3} \Rightarrow \int_{-\infty}^{+\infty} \frac{4\cos x}{(x^2+1)(x^2+9)} dx = \frac{2\pi}{4e} - \frac{2\pi}{12e^3}$$

且原函数为偶函数.

$$\therefore \int_0^{+\infty} \frac{4\cos x}{(x^2+1)(x^2+9)} dx = \frac{\pi}{4e} - \frac{\pi}{12e^3}$$

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11. 证: z_0 是 $f(z)$ 的 3 级极点

$\therefore z_0$ 是 $f'(z)$ 的 4 级极点

$\therefore z_0$ 是 $\frac{f'(z)}{f(z)}$ 的 $(4-3)$ 级极点 (1 级极点)

$$\therefore \operatorname{Res}\left(\frac{f'(z)}{f(z)}, z_0\right) = \lim_{z \rightarrow z_0} \frac{f'(z)}{f(z)(z-z_0)} = -3.$$

$$\therefore \operatorname{Res}\left(\frac{f'(z)}{f(z)}, z_0\right) = -3.$$

$$\text{令 } f(z) = \frac{1}{(z-z_0)^3}, \quad f'(z) = \frac{-3}{(z-z_0)^4}.$$

$$\therefore \frac{f'(z)}{f(z)} = \frac{-3}{(z-z_0)}.$$

$$\therefore \operatorname{Res}\left(\frac{-3}{(z-z_0)}, z_0\right) = \frac{-3}{(z-z_0)}(z-z_0) = -3.$$

$$\therefore \operatorname{Res}\left(\frac{f'(z)}{f(z)}, z_0\right) = -3.$$

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